
EMERGENT SUBSTRATE GRAVITY

Spin-2 Recovery of the Field Equations, the Thermodynamic Equation of State, and a Layered Falsification Ledger

RESEARCH ARTICLE • GRAVITATION AND FIELD THEORY

Foundations of the classical field equations and their falsifiable boundaries

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ABSTRACT. We present a self-contained account of classical gravity organized as two logically independent layers, with an explicit statement of which claims are forced, which are interpretive, and which are imported. In the first layer we take a symmetric rank-2 field $h_{\mu\nu}$ on a flat background and impose one requirement, linearized gauge invariance. This uniquely fixes the Fierz–Pauli Lagrangian, forces exactly two transverse-traceless (helicity ± 2) modes, and, through coupling to the full stress-energy tensor, fixes the post-Newtonian structure reproducing the classical solar-system observables. Requiring the field to source itself produces, at second order, the gravitational stress-energy with an algebraically fixed coefficient; we compute this explicitly, recover the Isaacson energy density $t_{00} = (c^4/32\pi G)\langle\dot{h}_{ij}\dot{h}^{ij}\rangle$, and verify the second-order Bianchi consistency symbolically. The all-orders closure to the Einstein–Hilbert action is the classic self-coupling theorem, cited rather than re-derived. In the second layer we record the sense in which these equilibrium field equations coincide with a thermodynamic equation of state $\delta Q = T \delta S$ across local horizons. We then separate a testable interpretive hypothesis, that the graviton is emergent (phonon-like) rather than fundamental, from the forced results, and state that current experiments do not settle it. We close with a falsification ledger whose tensor and thermodynamic layers fail independently, distinguishing algebraically forced “survivals” (which are the content of general relativity) from realizable death conditions.

KEYWORDS *gravitation • spin-2 field • Fierz–Pauli • self-coupling bootstrap • horizon thermodynamics • falsifiability*

1. INTRODUCTION AND SCOPE OF THE CLAIM

General relativity (GR) is the most precisely tested theory of gravity available, and nothing below challenges its empirical record. The question addressed here is organizational and interpretive: given that a spin-2 geometric carrier can be derived from symmetry alone, what does that derivation establish, and what does it leave open?

Two familiar facts sit in tension. First, a massless spin-2 field consistently coupled to matter reproduces GR, a classic result of field theory [1, 2, 3, 4, 5, 6, 7, 8, 9]. Second, the Einstein field equations follow from horizon thermodynamics

as an equation of state [11, 12]. It is tempting to read the first as mandating a *fundamental* graviton and the second as mandating an *emergent* gravity, and to treat them as competitors. We argue they are descriptions at different levels, and that the frequently asserted step from “spin-2 carrier” to “fundamental gauge particle” is an interpretive addition, not a forced conclusion.

We are explicit about the status of the contribution. The spin-2 recovery of GR is not new; we present it in self-contained form with an explicit second-order computation and a symbolic consistency check, flagging the one step (all-orders resummation) taken from the literature. The ther-

modynamic reading is Jacobson's, absorbed without extension. The new elements are: (i) the explicit separation of the tensor-recovery and thermodynamic layers into independent, separately falsifiable strata; (ii) a falsification ledger distinguishing algebraically forced results from realizable death conditions; and (iii) a precise typing of the emergent-versus-fundamental question as an interpretive hypothesis the present ledger cannot decide. Any novel empirical content of the wider program lies in its weak-field (galactic) sector, developed separately and not claimed here; the core discussed here *recovers* GR rather than modifying it.

2. FRAMEWORK: A CONTINUOUS SUBSTRATE AND THE TWO-LAYER METHOD

We adopt a continuous-field working picture: the gravitational degrees of freedom are the stress response of an underlying continuous medium rather than excitations on a passive background. We stress that this ontology is an *interpretation*. None of the derivations in Sections 3–5 depend on it; they depend only on the field content, the symmetry, and standard field theory. We keep the interpretive layer visibly separate from the forced layer, and do not treat the medium picture as evidence for any equation.

The analysis proceeds in two logically independent strata that, as shown in Section 11, fail independently: the **tensor-recovery layer** (Sections 3, 4, 7, 8), resting only on rank-2 tensor algebra, gauge invariance, and the Bianchi identities; and the **thermodynamic layer** (Sections 5, 9), resting on horizon entropy scaling and the Unruh relation.

3. FIRST-ORDER SECTOR: GAUGE INVARIANCE FORCES SPIN-2

Write $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $h_{\mu\nu}$ symmetric, signature $(-, +, +, +)$. Impose one requirement, invariance of the free action under

$$\delta h_{\mu\nu} = \partial_\mu \xi_\nu + \partial_\nu \xi_\mu. \quad (1)$$

A Lorentz-invariant, two-derivative Lagrangian quadratic in $h_{\mu\nu}$ and invariant under this transformation has its four kinetic terms fixed, up to overall normalization, to the Fierz–Pauli combination [1]; in the standard normalization reproducing

the Einstein coupling,

$$\mathcal{L}_2 = \frac{1}{2\kappa} \left[-\frac{1}{4} \partial_\lambda h_{\mu\nu} \partial^\lambda h^{\mu\nu} + \frac{1}{2} \partial_\lambda h^\lambda{}_\nu \partial_\mu h^{\mu\nu} - \frac{1}{2} \partial_\mu h^{\mu\nu} \partial_\nu h + \frac{1}{4} \partial_\lambda h \partial^\lambda h \right] + \frac{1}{2} h_{\mu\nu} T^{\mu\nu}, \quad (2)$$

with $\kappa = 8\pi G/c^4$. The relative coefficients of the four kinetic terms are what gauge invariance fixes; any other ratio propagates a ghost or a scalar mode. Gauge invariance of the source term requires $\partial_\mu T^{\mu\nu} = 0$. The equation of motion is $G_{\mu\nu}^{(1)} = \kappa T_{\mu\nu}$, which in Lorenz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$ becomes

$$\square \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu}, \quad \bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h. \quad (3)$$

Counting degrees of freedom, $10 - 4 - 4 = 2$: ten symmetric components, four Lorenz constraints, four residual gauge choices, leaving two transverse-traceless polarizations of helicity ± 2 . No scalar (breathing) or vector mode survives. Because the source is the full $T_{\mu\nu}$ rather than its trace, both h_{00} and h_{ij} carry the potential, fixing the parametrized post-Newtonian (PPN) parameter $\gamma = 1$ (Section 7). In the static, non-relativistic limit the 00 component reduces to $\nabla^2 \Phi = 4\pi G \rho$ with $h_{00} = -2\Phi/c^2$.

4. NONLINEAR COMPLETION: THE SELF-COUPLING BOOTSTRAP

The free field carries energy, and energy gravitates, so $h_{\mu\nu}$ must couple to its own stress-energy $t_{\mu\nu}$. Iterating this requirement is the classic bootstrap that promotes the linear theory to a nonlinear one [2, 3, 4, 5, 7, 8, 9]. We make the first iteration explicit, check its consistency, then cite the closure.

Defining the second-order Einstein tensor $G_{\mu\nu}^{(2)}$, the gravitational stress-energy is $t_{\mu\nu} \equiv -G_{\mu\nu}^{(2)}/\kappa$. Its coefficient is not free. The exact contracted Bianchi identity $\nabla^\mu G_{\mu\nu} \equiv 0$, expanded to second order, gives $\partial^\mu G_{\mu\nu}^{(1)} = 0$ at order h and $\partial^\mu G_{\mu\nu}^{(2)} = 0$ (terms $\propto G_{\mu\nu}^{(1)}$) at order h^2 . On shell ($G_{\mu\nu}^{(1)} = 0$) the second relation states $\partial^\mu t_{\mu\nu} = 0$: the graviton stress-energy is conserved, with a coefficient fixed entirely by the identity. We verified both relations symbolically for a fully generic $h_{\mu\nu}(t, x, y, z)$; each vanishes identically in all four components.

On a transverse-traceless wave $h_{xx} = -h_{yy} = p(t - z)$, $h_{xy} = q(t - z)$ (which satisfies $R_{\mu\nu}^{(1)} = 0$), the second-order Einstein tensor is, before averaging,

$$G_{00}^{(2)} = \frac{1}{2}(p')^2 + \frac{1}{2}(q')^2 + (p p'' + q q''). \quad (4)$$

The last two terms are total derivatives; wavelength-averaging, $\langle p p'' \rangle = -\langle (p')^2 \rangle$, yields

$$t_{00} = -\frac{1}{\kappa} \langle G_{00}^{(2)} \rangle = \frac{c^4}{32\pi G} \langle \dot{h}_{ij} \dot{h}^{ij} \rangle, \quad (5)$$

the Isaacson gravitational-wave energy density [10]. The coefficient emerges from the computation rather than being inserted. The remaining step, that iterating the self-coupling to all orders resums uniquely to the Einstein–Hilbert action $\sqrt{-g} R$ and hence $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$, is the self-coupling theorem [5, 7, 8, 9], taken from the literature and flagged as the one step not re-derived here.

5. MACROSCOPIC CLOSURE: THE THERMODYNAMIC EQUATION OF STATE

The tensor derivation yields the radiation and fluctuation sector. At the macroscopic, equilibrium level the same equations admit a thermodynamic reading. Following Jacobson [11], imposing the Clausius relation $\delta Q = T \delta S$ across every local Rindler horizon, with entropy proportional to horizon area [14, 15, 16] and temperature set by the Unruh relation [17], yields the Einstein field equations as an equation of state. Gravity’s equilibrium content is the metric configuration keeping the local heat flux consistent with the horizon entropy bound.

We absorb this result without extension, with two cautions in the open. First, the derivation is an *equilibrium* argument; an equation of state is not the full dynamics, just as a pressure–temperature relation is not the full hydrodynamics. Far-from-equilibrium dynamics are not established. Second, the entailment between the Clausius relation and the field equations runs both ways, so the derivation is *consistent with* emergence but does not prove it; a fundamental gravity whose horizon thermodynamics is a consequence fits the same mathematics. We therefore treat emergence as an interpretive hypothesis (Section 6), not a sealed result.

6. INTERPRETATION: EMERGENT VERSUS FUNDAMENTAL GRAVITON

The recovery of a spin-2 carrier does not settle whether the graviton is fundamental or emergent, and we do not claim it does. The interpretive hypothesis we record is that the graviton is an emergent collective excitation of the substrate, analogous to a phonon: quantized, but not fundamental. Under this reading the non-renormalizability of perturbatively quantized gravity is consistent with gravity being an effective field theory valid below a cutoff, rather than a defect requiring a specific ultraviolet completion. We emphasize what this does *not* establish. It does not prove a fundamental graviton is impossible; non-renormalizability is evidence of effective-theory status, not a proof of non-existence. The emergent and fundamental readings make identical predictions for every experiment considered here, and (Section 10) no experiment in our ledger distinguishes them. We carry the emergent-graviton claim as a hypothesis, explicitly not a derived conclusion.

7. CONSISTENCY FLOOR: CLASSICAL TESTS RECOVERED BY NECESSITY

The following are *not* predictions distinctive of the present interpretation. They are the forced content of *any* theory with the field content and symmetry of Section 3, i.e. the content of general relativity, recovered here by algebraic necessity. Predicted values combine the derivation with measured constants; confirmation values are current best measurements and should be re-checked before use.

- S1 Solar light deflection.** $1.751''$ at the limb, achromatic, PPN $\gamma = 1$. Full $T_{\mu\nu}$ coupling makes time and space curvature sum. Cassini: $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [21].
- S2 Mercury perihelion.** $+42.98''/\text{century}$, prograde, from $(2 + 2\gamma - \beta)/3 = 1$. A scalar coupling ($\gamma = -1$) gives a wrong-sign retrograde result.
- S3 GW polarization.** Two modes, helicity ± 2 , from $10 - 4 - 4 = 2$. Consistent with all reported LIGO–Virgo events.
- S4 Binary-pulsar decay.** Set by the Isaacson coefficient forced by the bootstrap. PSR B1913+16

consistent with GR to $\sim 0.2\%$ [22].

S5 Weak equivalence principle. $\eta = 0$; the test mass enters the action through a single coefficient. MICROSCOPE: $\eta = (-1.5 \pm 2.3 \pm 1.5) \times 10^{-15}$ [23].

S6 GW speed. $c_{\text{gw}} = c$ (massless). GW170817/GRB170817A: $-3 \times 10^{-15} < (c_{\text{gw}} - c)/c < 7 \times 10^{-16}$ [24].

8. FALSIFICATION I: THE TENSOR-SECTOR DEATH CONDITION

D1 (scalar or vector mode). *Falsified if* an interferometer network detects a scalar (breathing) or vector polarization component at $> 5\sigma$ in a confirmed event, showing the field is not governed purely by rank-2 gauge invariance and breaking the derivation of Section 3. D1 and S3 are the same measurement at its two edges: two modes is the survival, a third mode the death. Relevant instruments: Einstein Telescope, Cosmic Explorer, LISA.

9. FALSIFICATION II: THE EMERGENCE-SECTOR DEATH CONDITIONS

D2 (volumetric entropy scaling). *Falsified if* a gravitational or cosmological horizon is shown to carry entropy scaling with spatial volume, $S \propto V$, rather than boundary area, $S = A/4L_P^2$, removing the holographic input to the Clausius derivation.

D3 (equation-of-state decoupling). *Falsified if* a regime is found where the Einstein equations hold but horizon thermodynamics fails, or the reverse. The likely break point is far-from-equilibrium (early inflation, high-frequency radiation, singularity interiors), the open edge of Section 5.

10. FALSIFICATION III: QUANTUM DEGREES OF FREEDOM

Whether gravity carries quantum degrees of freedom is testable, within limits set by the Local Operations and Classical Communication (LOCC) theorem, through gravitationally induced entanglement [19, 20].

D4 (no quantum degrees of freedom). *Falsified if* a tabletop Bose–Marletto–Vedral experiment, two masses of order 10^{-14} kg in micron-scale spatial superposition, shows, *at the predicted sensitivity with decoherence systematics controlled*, that gravity

generates no entanglement, killing the quantized-excitation (phonon-like) stance.

Two clauses require care. *On a null result:* a null at predicted sensitivity with systematics controlled is *strong evidence* that gravity carries no quantum degrees of freedom, not a one-shot proof of strict classicality; absence of detected entanglement and proof of a classical channel are separated by the sensitivity and decoherence budget. *On detection:* it confirms the substrate carries quantized excitations but, by LOCC, cannot distinguish an emergent excitation (a gravitational phonon) from a fundamental gauge boson. Detection supports the quantized-excitation stance but leaves the emergent-versus-fundamental question open. Consequently that boundary is *not* falsifiable by any experiment in this ledger; we record it as a structural hypothesis, explicitly not a sealed result, so the ledger contains no death condition no instrument can meet.

11. DISCUSSION: LAYER INDEPENDENCE, PRIOR WORK, AND LIMITATIONS

Layer independence. The tensor conditions (D1, S1–S6) rely only on rank-2 tensor algebra and the Bianchi identities; the thermodynamic conditions (D2, D3) on entropy scaling and the Unruh relation. The geometric bootstrap can hold while volumetric-versus-area entropy scaling fails in an extreme regime, or the reverse. Separating the strata ensures a failure of the thermodynamic overlay does not register as a failure of the tensor algebra.

Relation to prior work. The spin-2 self-coupling recovery of GR is classical [2, 3, 4, 5, 7, 8, 9]; the thermodynamic derivation is Jacobson’s [11], within the broader emergent-gravity program [12, 13]. Independent constructions such as the entropic-action approach of Bianconi [18] produce modified field equations from a relative-entropy action; by contrast the core here recovers standard GR rather than modifying it, so we do not present it as a new predictive theory. Consensus among these approaches is treated as shared lineage, not evidence for any specific claim here.

Limitations. (i) The all-orders resummation is cited, not re-derived; only the first iteration and its coefficient are computed here. (ii) The thermodynamic derivation is an equilibrium result;

far-from-equilibrium dynamics are open (D4). (iii) The emergent-graviton interpretation is a hypothesis current experiment does not decide (Section 10). (iv) The continuous-substrate ontology is interpretive and carries no derivational weight. (v) Any novel empirical content of the wider program lies in its weak-field sector, developed elsewhere; the core here reproduces GR and inherits both its successes and its unsolved problems.

12. CONCLUSION

From a symmetric rank-2 field and a single symmetry we recover the spin-2 kinematics, the classical PPN observables, and, through self-coupling, the Einstein field equations, with the second-order gravitational stress-energy and its Isaacson coefficient computed explicitly and checked for Bianchi consistency. At the macroscopic level these equilibrium equations coincide with a thermodynamic equation of state. Forced results, interpretive hypotheses, and imported theorems are kept visibly distinct, and the falsification ledger's tensor and thermodynamic layers fail independently through realizable experiments. The one interpretive claim that motivated the framing, that the graviton is emergent rather than fundamental, is carried as a hypothesis current experiment does not settle. What is forced here is general relativity; what is proposed is a way of reading it; and the two are labeled as such throughout.

METHOD NOTE. This paper was developed under a private verification and organizing discipline that adds its cited results no warrant; every result rests solely on the physics references below, and the classical spine (Sections 3–5, 7) is standard field theory. Empirical confirmation values are current best measurements to be verified against the latest data.

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